

THE EFFECT OF U.S.–IRAN CONFLICT ON WTI AND BRENT CRUDE OIL PRICES: A TIME SERIES APPROACH



By

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BS STATISTICS

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A BS Project submitted to the department of Statistics (Data Science), Abdul Wali Khan University of Mardan (Garden Campus) In partial fulfilment of the requirement for the **Degree of BS Statistics (Data Science).**

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At any time, if my statement is found to be incorrect even after my graduation, the university has the right to withdraw m y BS degree.

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DEDICATION

*This project is dedicated to our Parents and to our professors **Dr Muhammad Aamir, Dr. Muhammad Hamraz, Dr. Nosheen Faiz, Dr. Umair Khalil** that has been a great source of motivation and inspiration for us, this is dedicated to all those who believe in themselves and the richness of learning.*

ACKNOWLEDGEMENT

First of all, we are very grateful Allah Almighty for blessing to the ability to complete this project, we would like to express our appreciation thanks to our supervisor Dr Muhammad Aamir (Associate Professor and Director of admissions), Prof. Dr Umair Khalil Chairman of Department of Statistics, that has always been a guide role for us, Dr Muhammad Hamraz Assistant Professor the honorable Personality that has always been with us leading us in hard time through his absolute greatness and Dr Nosheen Faiz Assistant Professor, that has been the great motivation and inspiration for us in these previous years, Laiba Alam a former senior student of English Department Abdul Wali Khan University Mardan Garden Campus, that made sure to teach us with all her might and build our foundation, Special thanks to our family members specially Parents that appreciated us and supported us through everything and made this possible for us providing us the facility that many of our younger generation lacks, Words cannot express how grateful we are to all of them for their different kind of support and prayers that made this journey possible for us.

Umer Riaz,

Manzoor Ali,

Muhammad Saeed,

Khayam Khan

ABSTRACT

The current study aims to analyze the effect of war (America-Iran) on Crude WTI oil and brent oil. The Researchers Group aims to analyze measurement by time series analysis. This was a 12-day conflict that began on 13 June 2025 when Israel launched large-scale airstrike on Iranian military and nuclear related targets. Iran responded with missile and drone attacks against Israel. On 22 June 2025, the United States entered the conflict directly by striking three Iranian nuclear sites: Fordow, Natanz, and Isfahan. Iran later launched missiles toward a U.S. base in Qatar. A ceasefire was reached on 24 June 2025. After 2025 conflict, On 28 February 2026, the U.S. and Israel launched a much broader military campaign against Iran, Iran responded with attacks on U.S. military and diplomatic targets across regions and disruption around the Strait of Hormuz. That disruption caused a partial blockage for oil supplies to move through the Persian Gulf to the rest of the world. That effected highly in their changes throughout the globe. This study aims to analyze all the changes that this disruption caused to Oil market overall from 3 months before the conflict and after the conflict, By ARIMA model we would predict and analyze the changes in oil before and after 28 February 2026. This analysis will be helpful to find out the effect of War (America-Iran) on Oil resources and forecast the upcoming changes throughout the world.

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LIST OF ABBREVIATIONS

WTI	West Texas Intermediate
AC	Autocorrelation
ARIMA	Auto regressive integrated moving average
US	united states
UK	United Kingdom
AR	Auto regressive/regression
I	Integrated/Integration
MA	Moving average

CHAPTER 1

INTRODUCTION

1.1 Introduction

This research will analyze the effect of IranUS conflict on the WTI (West Texas Intermediate) and Brent crude oil prices. The occurrence of geopolitical conflict may influence the world oil market through uncertainties that might result into fluctuations in oil prices.

1.2 USIRAN Conflict History

The United States and Iran have been against each other from Aug 1953 when U.S. helped Iran the overthrow of Mohammed Mossadegh (the democratic prime minister of Iran. Then came the proclaimed Crown Prince also known as Mohammed Reza Shah Pahlavi (the last of Shah), To whom the U.S. showed support. whose security services brutalized Iranian citizens for decades. The Shah's Rule was increasingly autocratic. Political dissent was suppressed through the secret police, SAVAK and opposition from religious leaders, Leftist groups and student grew by 1978 and took over the U.S. Embassy in Tehran in November 1979, resulting in economic sanctions and the severing of formal diplomatic relations between the nations. Since 1984, the U.S. State Department has listed Iran as a "state sponsor of terrorism," alleging the Iranian government provides terrorists with training, money and weapons. Some of the major events in U.S.Iran relations highlight the differences between the nations' views, but others arguably presented real opportunities for reconciliation.

The Iranian Parliament appointed Mohammad Mossadegh as prime minister In 1951, Mossadegh guided legislators to nationalize the Anglo Iranian Oil Company, remove its British proprietors and allocate oil revenues toward investments for Iranian populace. The United States expressed concerns over the potential disruptions to the global oil supply and the prospect of increased Soviet influence in Iran, while the United Kingdom was apprehensive about losing access to low-cost Iranian Oil.

President Dwight Eisenhower concluded that it was in the best interests of the United States and the United Kingdom to remove Mossadegh from power. Operation Ajax, a joint CIA British mission, persuaded the Shah of Iran to depose Mossadegh and remove him by force. Mossadegh was then replaced by a more pro-Western prime minister chosen with CIA support. After more than 25 years of relative stability in U.S.-Iran relations, public dissatisfaction in Iran had increased because of the social and economic conditions that developed under the authoritarian rule of Shah Mohammad Reza Pahlavi.

Pahlavi accumulated wealth and used American assistance to strengthen the military, while many Iranians lived in poverty. Dissent was frequently suppressed violently by SAVAK, the shah's security service. In January 1979, the shah left Iran, reportedly for cancer treatment. Two weeks later, Ayatollah Ruhollah Khomeini returned from exile in Iraq and led efforts to abolish the monarchy and establish an Islamic government.

In October 1979, President Jimmy Carter agreed to let the shah enter the United States for advanced medical treatment. In response, outraged Iranian students seized the U.S. Embassy in Tehran on Nov. 4 and took 52 Americans hostage. That action led Carter to cut U.S. diplomatic relations with Iran on April 7, 1980. Two weeks later, the U.S. military attempted a rescue mission, but it failed after aircraft crashes killed eight U.S. servicemembers. The shah died in Egypt in July 1980, but the hostages were not released until January 20, 1981, after 444 days of captivity. In August 2002, an exiled rebel group announced that Iran had been secretly developing nuclear weapons at two facilities that had not previously been disclosed publicly. This violated the terms of the Nuclear Nonproliferation Treaty, which Iran had signed and which requires countries to report their nuclear-related facilities to international inspectors. One of those previously secret sites, Natanz, contained centrifuges used to enrich uranium, which can be used in civilian nuclear reactors or enriched further for weapons. Beginning around 2005, U.S. and Israeli government cyberattacks reportedly targeted the Natanz centrifuges with a custommade piece of malicious software known as Stuxnet. This operation, which slowed Iran's nuclear program, was one of many U.S. and international efforts, most of which were unsuccessful, to prevent Iran from developing a nuclear bomb.

After a decade of unsuccessful efforts to curb Iran's nuclear ambitions, the Obama administration adopted a direct diplomatic strategy in 2013. Two years of secret direct negotiations, initially between the United States and Iran and later expanded to include other nuclear powers, led to the Joint Comprehensive Plan of Action, commonly known as the Iran nuclear deal. Iran, the United States, China, France, Germany, Russia, and the United Kingdom signed the agreement in 2015. The deal significantly limited Iran's capacity to enrich uranium and required international inspectors to monitor and verify Iran's compliance. In return, Iran received relief from international and U.S. economic sanctions. Although inspectors regularly certified that Iran was meeting the agreement's terms, President Donald Trump withdrew from the deal in May 2018.

On January 3, 2020, an American drone fired a missile that killed Major General Qassem Soleimani, the commander of Iran's elite Quds Force. Analysts regarded Soleimani as the second most powerful figure in Iran, after Supreme Leader Ayatollah Khamenei. At the time, the Trump administration claimed that Soleimani was planning an imminent attack against U.S. assets in the region, though officials did not provide clear evidence to substantiate that claim. Iran retaliated by launching ballistic missiles at two American bases in Iraq.

Hamas's attack on Israel on October 7, 2023, triggered a forceful military response from Israel that continues to this day and significantly weakened Iran's regional proxies, particularly Hamas, the group responsible for the assault, and Hezbollah in Lebanon. In early 2026, successive rounds of indirect negotiations were held between Iran and representatives of the U.S. administration.

These talks followed major unrest in Iran, during which Trump told protesters that "help is on its way." Then, on February 28, the United States and Israel began bombing Iran in an operation the United States called "Epic Fury." In the initial wave of airstrikes, Supreme Leader Ali Khamenei and other senior figures of the Islamic Republic were killed. Tehran responded by striking targets across the Gulf, widening the conflict into a broader regional confrontation.

1.3 Background of the study

This clash between Iran and US has escalated into a military conflict by late February 2026, which resulted in disturbance in international oil markets owing to partial blockade of the Strait of Hormuz. This led to the increase in WTI oil price from \$62.18 to \$96.47 per barrel for the period of 12/8/2025 to 3/6/2026. On the other hand, the rise in Brent oil price is from \$66.63 to \$101.52. (Investing.com, 2026)

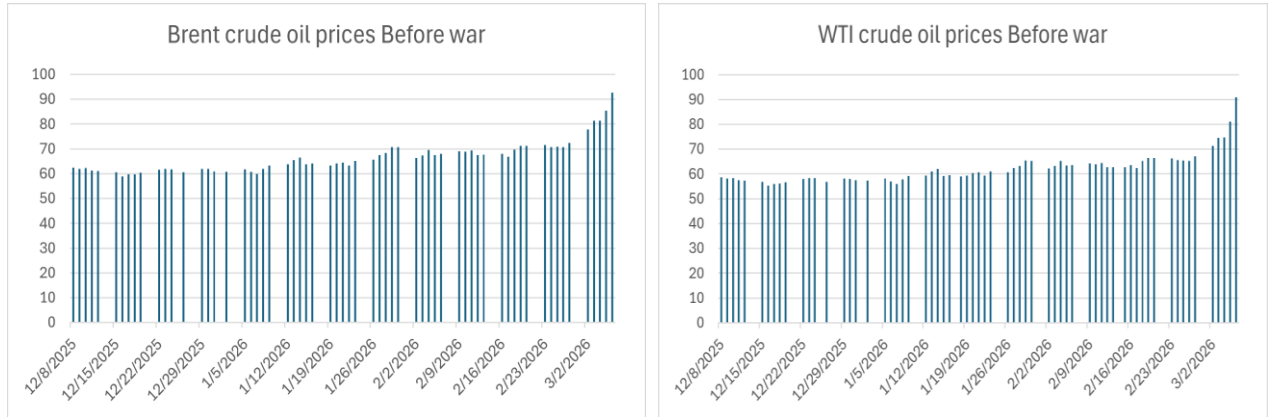


Figure 1.1: WTI crude oil prices, December 2025 – March 2026 (pre-conflict period) and WTI crude oil prices, March 2026 – June 2026 (post-conflict period)

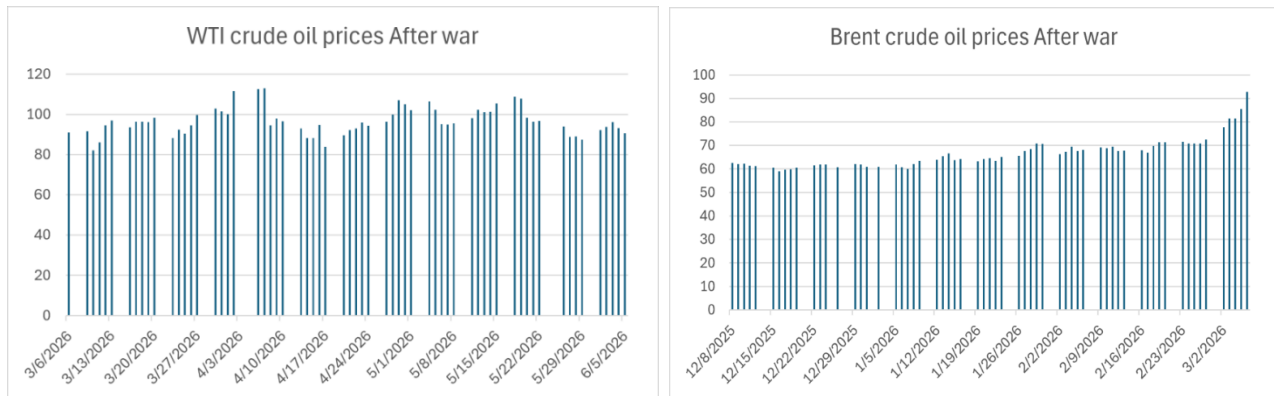


Figure 1.2: Brent crude oil prices, December 2025 – March 2026 (pre-conflict period) and Brent crude oil prices, March 2026 – June 2026 (post-conflict period)

1.4 Problem Statement

Oil prices are affected by a lot of things including what's happening in the economy and in politics. One big event that has happened recently is the problem between Iran and the United States. When things like this happen it can make people unsure about what will happen in the oil market. That can make the prices of WTI and Brent crude oil go up or down. Nobody has really looked at how this problem between Iran and the United States affected the prices of WTI and Brent crude oil using numbers and looking at what happened over time.

This study is going to look at what happened to the prices of WTI and Brent oil before and after the problem between Iran and the United States started. We will use numbers to see how the prices changed. We will use a special tool called the ARIMA model to try to guess what the prices will be in the future based on what happened in the past. By doing this study we can get an understanding of how things like the problem between Iran and the United States can affect oil prices and what might happen to them in the future. Oil prices and the conflict between Iran and the United States are very important to understand. This study will help us learn more about oil prices and how they are affected by things like the conflict, between Iran and the United States.

1.5 Objective of the study

The main objectives of this study are:

- i. To analyze and forecast Brent WTI and Crude Oil prices
- ii. To analyze and forecast Brent WTI and Crude Oil prices before and after war.

1.6 Significance of the study

This study is highly important for academic research as well as for real-world forecasting in finance and energy economics. It improves our understanding of traditional, hybrid, and ensemble forecasting methods for crude and brent oil prices by comparing their accuracy. From an academic point of view, the findings contribute to the existing study by examining different forecasting methods within a single empirical setting. In addition, the study provides methodological insights that can support future research in timeseries forecasting and ensemble modelling.

More specifically, traders, governments, energy providers, and other market participants involved in risk management, strategy development, and investment decisions can benefit from more accurate crude and brent oil price forecasts. The proposed forecasting method can help participants predict price changes more precisely and respond quickly to market instability. Overall, this research offers an effective and practical approach to WTI crude and brent oil price forecasting that can also be applied to other commodities and economic time series, thereby improving forecasting methods across various fields.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews the present literature associated to the forecasting of crude and brent oil prices and shows the important methods used in previous studies. The crude and brent oil are one of the main trading commodities and its price fluctuations has significant effect on financial and economic activities, inflation rate and energy security strategies [2]. Change in both crude and brent oil have shown significant effect on both developed and underdeveloped countries economy through manufacturing expenses and balance trade [3]

Recent proofs proposed that the prices of crude and brent oil are very sensitive to political instability and global conflicts [4] supply problem based on production reductions and transportation limitation have also been related to high fluctuations in oil prices [5]

Global financial crisis and worldwide recessions have increased the volatility of oil prices, raising market uncertainty [6] by result, forecasting of Crude and brent oil prices accuracy has become an important point for traders, energy developing organizations and policymakers [7] The wider range of forecasting reaches has been suggested in the thesis.

That may range from classical statistical models to developed hybrid and ensembled techniques. these time series and ensemble model purpose to handle volatility and complex dynamics in the series of Crude and brent oil prices [8] This chapter focuses on time series models, volatility-based methods, state space approaches, and decomposition-based ensemble forecasting methods.

2.1 Time Series Forecasting in Crude and Brent Oil

Time series forecasting has been crucially used in crude and brent oil markets because past price data have very important information about future price fluctuations [9]. The dynamic pattern of crude and brent oil prices used econometric models to identify trends and cyclical behavior [10]. Good forecasts help trends in create decisions about treading and assets allocation [11]. Government and regulatory organizations also depend on oil price forecasts to design financial and energy strategies [12]

2.2 ARIMA Model

The autoregressive integrated moving average model, also known as ARIMA(p,d,q): the dependent variable must be stable (via the I component), and the independent variables are defined as all lags of the dependent variable (the AR component) and error lags. In General, one may think of an ARIMA model as a particular kind of regression model (Because of the MA component), p represents the AR coefficient and the constant term, while q represents the MA coefficient.[13].

Given the significant influence of oil price fluctuations on inflation, investment, and financial markets, forecasting crude and brent oil prices remain a major challenge in energy economics, risk management, and macro financial stability. Traditional univariate autoregressive integrated moving average (ARIMA) models are frequently utilized for crude and brent oil price forecasting and provide a good basis for short term predictions [14]. The ARIMA model forecasts values based on their linear depending on previous values. The concepts mentioned in previous sections apply to stationary data series. Most time series data are nonstationary. To fit stationary models, it is important to remove the fluctuation of nonstationary sources in time series data.[15]. Due to their simple and flexibility, ARIMA models are often used as statistical forecasting tools in energy economics research [16].

Empirical studies show that ARIMA model applies well for short-term crude oil price prediction under stable market conditions [17]. However, ARIMA models assume linear dynamics and constant variance, which reduce their usefulness during periods of market volatility The ARIMA model does not effectively account for the volatility in the data [18]. Quick financial breakdowns and big price shocks its significantly reduced by ARIMA forecasting accuracy, because its model do not have handle high shocks in price it is very use full for short term data [19].

2.3 Summary

In summary the forecasting of crude and brent oil prices is not an easy task, but previous research gives significant insights and is used as a basis for the development of new forecasting techniques.

CHAPTER 3

STATSTICAL ANALYSIS AND RESULTS

3.1 Introduction

In this chapter, the statistical analysis of WTI and Brent crude oil prices before and after the outbreak of the conflict between Iran and the US is performed. In order to find out whether there was any statistically significant difference between the levels of prices, their dispersion, and the degree of condonement before and after the outbreak of the conflict, descriptive statistics, correlation analysis, and hypothesis testing (Welch's t-test and the Mann-Whitney U test) were done using R.

3.2 Descriptive Statistics

Table 1.1 Presents the mean, variance and standard deviation of WTI and Brent Crude Oil Prices for the period before and after the conflict.

Table 3.1: Descriptive Statistics of WTI and Brent Crude Oil Prices

Statistics	WTI Before	WTI After	Brent Before	Brent After
Mean	62.18	96.47	66.64	101.52
Variance	51.57	45.08	41.72	51.57
Standard deviation	6.05	6.71	6.46	7.18

It can be observed that the average price increased significantly for both the oil benchmarks after the occurrence of the war, with WTI increasing by about \$34.29 per barrel and Brent increasing by about \$34.89 per barrel. The standard deviation was also higher for both the oil benchmarks after the outbreak of the war.

3.3 Correlation Analysis

To assess the extent of the two benchmarks' movement together, the Pearson correlation coefficient between WTI and Brent prices was calculated separately for the before and after periods using R

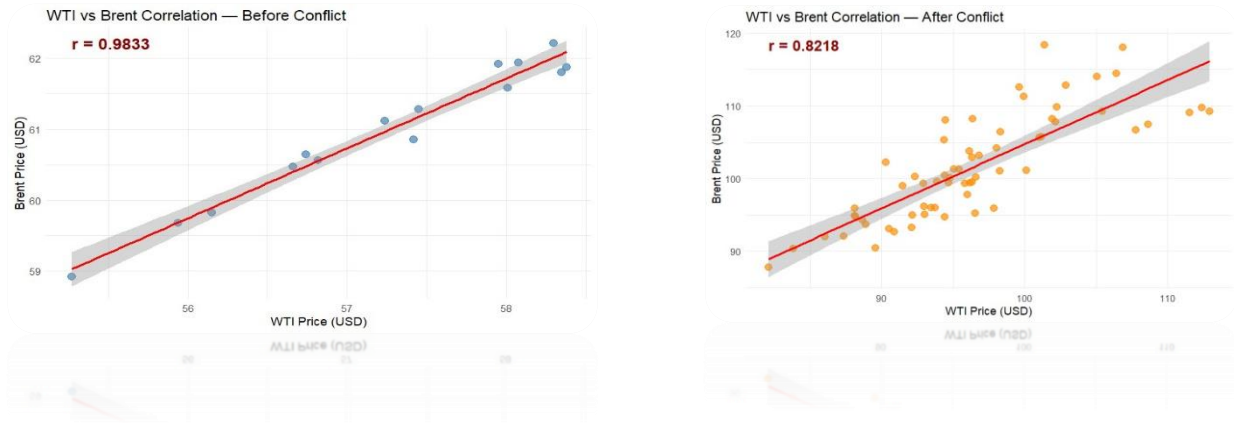


Figure 3.1: WTI vs Brent Correlation – Before conflict and WTI vs Brent Correlation – After conflict

Table 3.2: Correlation Between WTI and Brent Oil Prices

Period	Correlation Coefficient (r)
Before Conflict	0.9833
After Conflict	0.8218



Figure 3.2: WTI vs Brent Correlation – Before & After conflict (Combined)

This study shows that before the conflict WTI oil prices and Brent oil prices were very close they usually went up and down together. The connection between WTI oil prices and Brent oil prices was very strong. This means that WTI oil prices and Brent oil prices generally followed the trends in the market. After the conflict started WTI oil prices and Brent oil prices did not move closely together the connection between WTI oil prices and Brent oil prices was not as strong as before.

One reason for this is that Brent oil prices are more affected by things that happen in the Middle East and by problems with getting oil to places. On the hand WTI oil prices are more influenced by how much oil is produced in the United States and how much oil is stored there.

3.4 Welch Two-sample t-test and Mann-Whitney U test

3.4.1 WTI Crude Oil price

➤ These are the results of T-Test

A Welch two-sample t-test was carried out in order to test whether there is a statistically significant difference between Brent crude oil prices before and after the IranUS dispute.

- $t = 28.916$
- $p\text{-value} < 2.2e16$
- Mean before = \$66.63508
- Mean after = \$101.52262
- 95% CI = 37.27 to 32.49

Discussion

Like WTI, there was a significant increase in Brent crude oil prices in the aftermath of the IranUS dispute. The mean price increased from \$66.63 (before the dispute) to \$101.52 (after the dispute), meaning that the prices went up by approximately \$35 per barrel, $t(125.31) = 28.9$

➤ Results for Mann-Whitney U test

- $W = 12.5$

- $P\text{-value} < 2.2e16$

Discussion

Prices before the conflict were rarely higher than after the conflict. This proves beyond reasonable doubt that there was a rise in the price of WTI after the conflict, just like in the T-test results.

3.4.2 Brent Crude Oil price

➤ Results of T-Test

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Discussion

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➤ Results of Mann-Whitney U Test

- $W = 5.5$
- $P\text{-value} < 2.2e16$

Discussion

Like WTI, the result of Mann-Whitney test is consistent with the earlier finding, where it can be seen that Brent oil prices were significantly higher post-conflict than pre-conflict.

3.5 Summary

In conclusion, there was evidence of a statistically significant rise in price for both oil benchmarks after the emergence of the conflict between Iran and the US, as verified using the t-test and Mann-Whitney U test.

CHAPTER 4

ARIMA TIME SERIES MODELING

4.1 Introduction

Chapter 3 showed that WTI and Brent oil prices changed a lot after the IranUS conflict started. The t-test, Mann-Whitney test, and correlation test all looked at average prices and how the two oils moved together. But these tests only look at one snapshot in time, they do not show how prices change day by day. This chapter uses a method called ARIMA to look at that. ARIMA is a way to study prices using only their own past values. It helps us see if the conflict changed not just the price level, but also the way prices move over time.

4.2 Methodology

An ARIMA model has three numbers: (p, d, q).

p = how many past prices the model looks at

d = how many times the data had to be "differenced" (a way to remove trend) before it became steady

q = how many past errors the model looks at

In R, the function `auto.arima()` was used to find the best (p, d, q) numbers automatically. It tries many options and picks the one with the lowest AIC score (a number that tells you how good the model is — lower is better). Six models were made in total: WTI (overall, before, after) and Brent (overall, before, after). After fitting each model, a test called the Ljung-Box test was used to check if the model worked well.

4.3 Data Collection

The first step in the research methodology is to collect the historical price data of crude oil, as this study is about predicting weekly crude oil prices. A thorough dataset covering a particular period of time is needed for training and analyzing forecasting models. This study will use weekly crude oil price data

from a financial database. The dataset will consist of three global marketplaces, which include Brent, WTI, and Dubai. The data will be split into two phases: testing and training. 80% of the data will be used to train models, with the remaining 20% for model analysis and forecasting. The research will use a multistep ahead forecasting method.

4.3.1 ARIMA (Auto Regressive Integrated Moving Average) Model

The Autoregressive Integrated Moving Average (ARIMA) model is one of the most used time series forecasting approaches. ARIMA is appropriate for nonstationary time series because it combines autoregression (AR), differencing (I), and moving average (MA) components to model both linear dependency and prior shocks in the data [67]. Crude oil prices, similar to many other financial time series, commonly show non-stationarity, trends, and serial correlation, making ARIMA an excellent basic forecasting method. The ARIMA model identifies linear patterns.

Autoregression + Differencing + Moving average

AR + I + MA

A. representation of the AR(p) model equation is as follows:

$$Y_t = C + \sum_{i=1}^p \alpha_i Y_{t-i} + \varepsilon_t$$

In this Model

- Y_t = The value of variable at the current time t .
- C = A constant (intercept)
- p = The order of the autoregressive model.
- α_i = Autoregressive coefficient
- Y_{t-i} = The value of the series i th time period before the current time.
- ε_t = random error

If X_t is Y_t 's differenced series, an ARIMA framework with orders p, d, q may be stated as:

$$\begin{aligned} X_t &= Y_t - Y_{t-1} \\ &= (1-B) Y_t \end{aligned}$$

B. backward shift operator

The MA component denotes to the dependencies of an observation on an error residual term from a moving average model used for lagged data. $MA(q)$ is a moving average approach of order q , where q represents the number of lagged forecast errors in the forecasting equation. The equation for an $MA(q)$ model is:

$$y_t = C + \sum_{j=1}^q \beta_j \varepsilon_{t-j} + \varepsilon_t$$

In this Model

- Y_t = The current value of time series at time t .
- C = A constant (Intercept)
- q = The order of moving average model.
- β_j = Moving average coefficient.
- ε_{t-j} = Previous error at lag j
- ε_t = Current random error

By combination of these terms, an ARIMA model is characterized by parameters have three: p , d , and q , which are represented as ARIMA (p, d, q). The ARIMA (p, d , and q) model complete equation is:

$$y_t = C + \sum_{i=1}^p \alpha_i y_{t-i} - (1-B)^d \varepsilon_t + \sum_{j=1}^q \beta_j \varepsilon_{t-j}$$

- Y_t = The value of the time series at the current time t .
- C = Contant (intercept)
- $\sum_{i=1}^p \alpha_i Y_{t-i}$ = This is the Autoregressive (AR) component
- $(1 - B)^d$ = This is the Integrated (I) component
- $\sum_{j=1}^q \beta_j \varepsilon_{t-j}$ = This is moving average (MA) component
- ε_t = This is random error

$$\phi(B)(1 - B)^d Y_t = \theta(B)\varepsilon_t$$

In this formulation, B represents the backward shift (lag) operator, which is represented by $t=y_{t-1}$. The polynomial $\phi(B)$ shows the autoregressive (AR) component, which explains the dependency of the series' current value on past data. $\theta(B)$ shows the moving average (MA) polynomial, which analyses the influence of past error terms on the current observation. The word ε_t refers to a white-noise error process with zero mean and constant variance, showing that the residuals are uncorrelated with each other. The ARIMA model combines these components to explain the linear structure of a stationary time series (after differencing, if needed), and is therefore widely used as an initial model against which advanced and hybrid forecasting models are compared [15].

4.3.2 Identifying models with ARIMA order selection

In ARIMA models, model identification is achieved by order selection. Order selection refers to selecting the appropriate values for the ARIMA parameters p , d , and q considering the time series data properties. There are many approaches for selecting the order of ARIMA.

4.4 ARIMA Model Results

Table 4.1 shows the model picked for each of the six price series, along with its numbers.

Table 4.1: ARIMA Model Results for WTI and Brent

Series	Model	Coefficient(s)	AIC	BIC	RMSE
WTI – Overall	ARIMA (1,0,1)		701.62	704.48	3.71
WTI – Before	ARIMA (0,2,1)	MA1 = 0.636 (SE = 0.121)	254.83	259.12	1.74
WTI – After	ARIMA (1,0,0)	AR1 = 0.727 (SE = 0.084); Mean = 96.03 (SE = 2.01)	382.56	389.04	4.56
Brent – Overall	ARIMA (0,1,1)	MA1 = 0.175 (SE = 0.090)	707.81	713.50	3.85
Brent – Before	ARIMA (0,2,1)	MA1 = 0.774 (SE = 0.099)	243.17	247.39	1.68
Brent – After	ARIMA (1,0,0)	AR1 = 0.736 (SE = 0.084); Mean = 100.84 (SE = 2.21)	397.01	403.53	4.87

A clear pattern shows up here. Before the conflict, both WTI and Brent needed the data to be differenced twice ($d = 2$) to remove a strong upward trend. After the conflict, neither series needed any differencing ($d = 0$) — instead, prices just moved around a steady average.

Figure 4.1: WTI – Before – Forecast, WTI – After – Forecast, and WTI – Overall
(Before and after) – Forecast

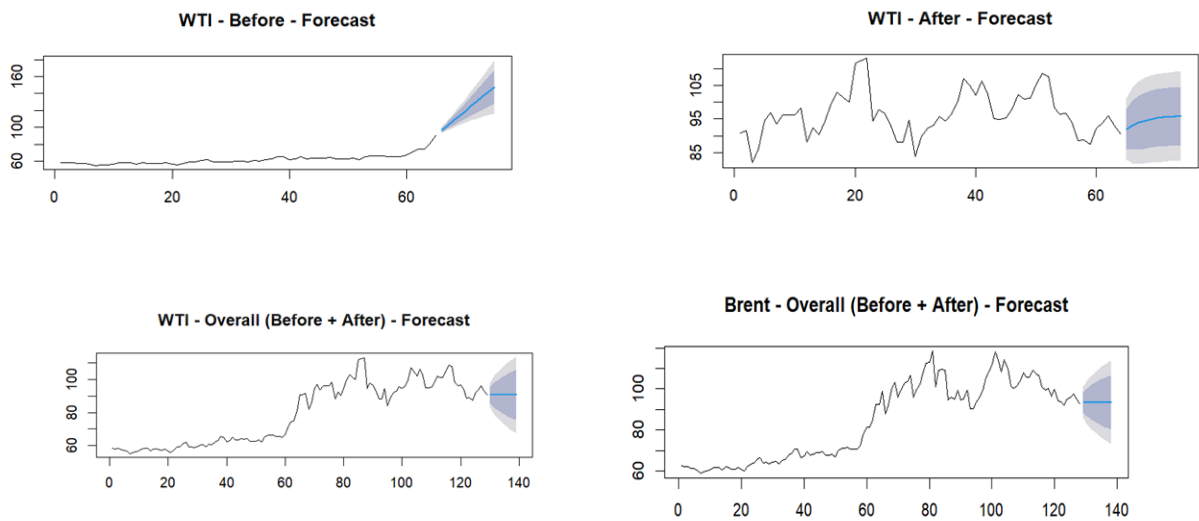


Figure 4.2: Brent – Overall (Before + After) – Forecast

4.5 Residual Diagnostics

After fitting each model, there is always a small gap between what the model predicted and the real price. This leftover gap is called a **residual**. A good model should leave residuals that look random, with no pattern left. The Ljung Box test checks this. Table 4.2 shows the results.

Table 4.2: Ljung Box Test Results

Series	Q*	df	pvalue	Residuals, OK?
WTI – Overall	23.45	10	0.0092	No
WTI – Before	7.80	9	0.5549	Yes
WTI – After	13.76	9	0.1311	Yes
Brent – Overall	10.17	9	0.3371	Yes
Brent – Before	11.79	9	0.2251	Yes
Brent – After	8.24	9	0.5104	Yes

A pvalue higher than 0.05 shows that the residuals are random; hence, the model is good. All models have satisfied this condition except for WTI Overall, whose pvalue is 0.0092. This indicates that one single model cannot appropriately represent the WTI price in the entire period but can do so if divided into two periods, which is precisely the approach used in this research.

4.6 Stationarity and Structural Change

The "d" value in each model tells us if the price data was steady (stationary) or not. Table 4.3 shows this clearly.

Table 4.3: Checking Stationarity

Series	d value	What it means
WTI – Before	2	Not steady, strong upward trend
WTI – After	0	Steady, moves around a fixed average
Brent – Before	2	Not steady, strong upward trend
Brent – After	0	Steady, moves around a fixed average

Before the conflict, both oils had a trend that kept climbing, even speeding up — that's what $d = 2$ means. After the conflict, both oils stopped trending and instead settled around a new, higher average price (about \$96 for WTI and \$101 for Brent).

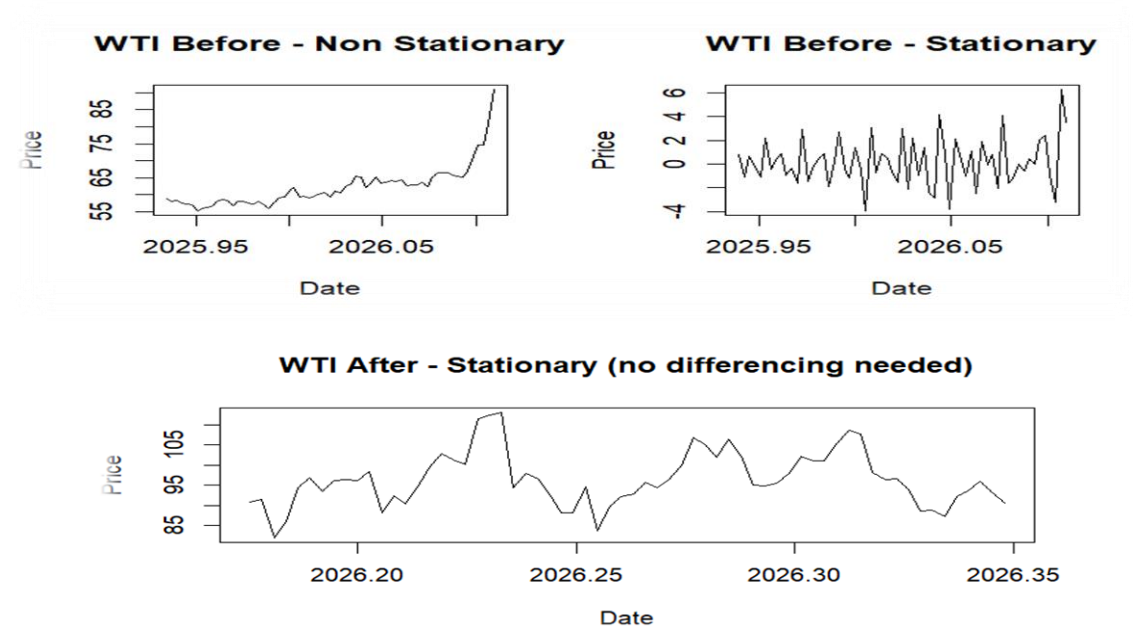


Figure 4.3: WTI Before conflict (non-stationary and stationary), WTI After – Stationary (no difference needed)

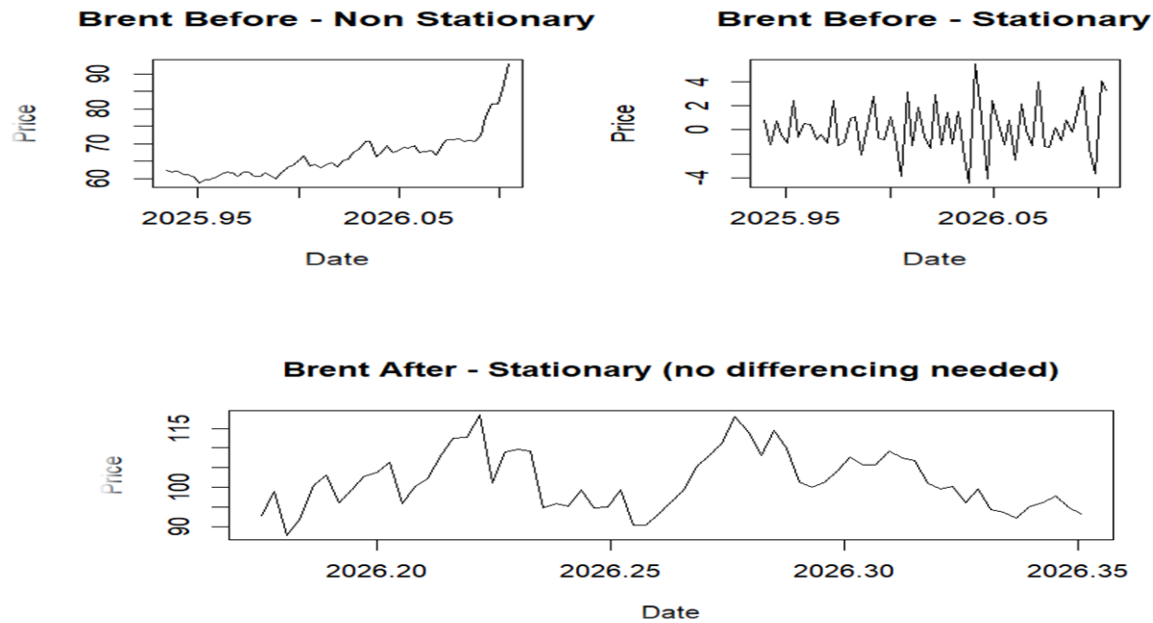


Figure 4.4: Brent Before conflict (non-stationary and stationary), Brent After – Stationary (no difference needed)

Before the conflict, both WTI and Brent showed a definite upward trend which became stationary after second order differencing ($d=2$). Post conflict, however, both time series were already stationary ($d=0$) and oscillated about their new, higher means rather than trending. This further reinforces the more general finding of this chapter: the oil price conflict affected not just the level of the prices but fundamentally changed the nature of their underlying process from an ascending trend to a stationary one centered around a higher equilibrium level (approx. \$96 for WTI and approx. \$101 for Brent). WTI and Brent also had similar models post conflict (ARIMA (1,0,0) for both, with similar coefficients of 0.727 and 0.736 respectively), indicating that the two oil types started acting similarly after reaching their new, higher equilibrium levels.

4.3 Chapter Summary

ARIMA models have been applied to analyses WTI and Brent oil prices prior to and during the IranUS conflict in this chapter. The results obtained reveal that there were changes in the pattern in which the prices changed due to the occurrence of the conflict. The prices of both the WTI and Brent exhibited a continuous rising trend prior to the conflict but once the conflict began, both prices did not exhibit any trends anymore and instead exhibited a random pattern about an elevated mean level. Graphical representation is provided in Figures 4.74.10 while numerical representation is provided in Table 4.3. This finding corroborates the results obtained in Chapter 4.

CHAPTER 5

CONCLUSION

5.1 Introduction

This chapter brings together the results of the previous chapters in final synthesis. Chapter 1 introduced the background of IranUS conflict and its impact on the global crude oil markets. In chapter 3, we used descriptive statistics, correlation analysis and hypothesis testing to compare WTI and Brent crude oil prices before and after the conflict. In Chapter 4, we used ARIMA time series modeling to explore the underlying dynamics of prices for both benchmarks over the same periods. This chapter presents a general overview of the findings, conclusions in relation to the objectives of the study, practical implications, limitations of the study and directions for future research.

5.2 Thesis Summary

This research sought to find out whether the outbreak of the Iran-U.S. war in late February 2026 had any significant impact on the price level, volatility, and condonement of WTI and Brent crude oil prices and whether the time series properties of the two benchmarks have been altered due to the event.

Descriptive statistics revealed that average WTI prices increased from \$62.18 per barrel prior to the war to \$96.47 per barrel after the war. Likewise, average Brent prices increased from \$66.64 to \$101.52. There were statistically significant increases in both benchmarks, as verified by the Welch two-sample t-test and Mann-Whitney U-test ($p < .001$). Correlation analysis revealed that the correlation between WTI and Brent prices before the war was very high ($r = 0.9833$), while the condonement became much lower in the case of the war ($r = 0.8218$).

The ARIMA model considerably enhanced these results. Prior to the war, the WTI and Brent were exchanged in a nonstationary, trending manner. To make the series stationary, a second degree of differencing ($d = 2$) was needed. But instead of following a straightforward autoregressive process fluctuating around a new, higher mean, both series were completely stationary ($d = 0$) following the fight. Both the before and after-conflict

models were well specified, according to residual diagnostics (Ljung-Box test). The choice to examine the two periods independently throughout this investigation was further supported by the fact that a single model fitted to the entire combined period failed the test.

5.3 Conclusion

The study's conclusions demonstrate that the IranUS dispute had a statistically significant impact on global crude oil prices. Over time, the battle changed the behavior of crude oil in addition to raising its price. WTI and Brent prices were steadily increasing prior to the dispute. Both benchmarks underwent a shift to a steady mean reverting process around a permanently high price level after the war.

This suggests that rather than a brief price hike, the fighting caused a structural disruption in the crude oil market. These results are congruent with previous research conducted on important geopolitical incidents like the wars in the Persian Gulf, Iraq, and Russia/Ukraine, which revealed similar changes in oil prices in the long run.

The WTI and Brent correlation is not as strong after the conflict. This shows that even though WTI and Brent were both affected Brent was a bit more affected because of the conflict near the Strait of Hormuz. Brent is closely linked to the Middle East in terms of location and trade, which is why it was more vulnerable than WTI. WTI and Brent are both important. Brent is more connected to the Middle East and that is why Brent was more affected by the conflict. The impact of the conflict on WTI and Brent was different because of where they're located and how they are used in trade.

5.4 Practical Implications

For people who make energy policies and regulators the big jump in oil prices after the conflict shows that global energy markets are still very sensitive to problems in areas like the Strait of Hormuz. This means that having oil reserves and different sources of energy can help protect against future shocks caused by geopolitics.

- The Ljung-Box test for the WTI Overall model shows that one ARIMA model is not enough to explain the period before and after the conflict. This tells us that forecasting

models should consider changes around major events, like conflicts instead of assuming that oil prices follow a simple and steady path.

5.5 Limitations

This research has some limitations like any study.

1. **Data source:** We got our price data from one place, Investing.com. It's an trustworthy site for old commodity prices but we didn't check it against other sources like EIA or Bloomberg.
2. **Sample size and Time Frame:** We collected data for about three months before and after a conflict. Our time frame after the conflict is long enough to use some methods but a longer one would help us see if the new price level really sticks or if prices go back to normal over time. We chose our model using a method called `auto.arima()` which picks the best model based on something called AIC
3. **Model selection:** This approach is okay. It might not always pick the very best model. We didn't try types of models like ones that account for volatility. We tried to isolate the effect of the conflict on oil prices by comparing after but we didn't account for other things that could have affected prices.
4. **Exogenous factors:** Things like production decisions changes in global demand or other big economic events might have played a role in the price changes we saw. So some of the changes might not be due, to the conflict itself.

5.6 Recommendations for Future Work

1. **Make the Study Longer:** Future studies should look at what happens after a conflict for a time. This will help us see if the higher price level and steady behavior keep going or go back to how things were
2. **Measure of Changes:** Researchers should use special models like GARCH or EGARCH. These will help them understand and predict how much oil prices can change during conflicts for WTI and Brent crude oil.

3. **Consider Factors:** Future research should include things like OPEC+ production decisions the US dollar index and global demand. They can use models like ARIMAX or VAR for this. This way they can separate the impact of the IranUS conflict from things that might affect the market.
4. **Check for Stationarity:** Future studies should do tests like ADF, KPSS and Phillips-Perron. These tests will help confirm if the data's steady or not.
5. **Compare with Conflicts:** Looking at other historical conflicts like the Persian Gulf War, Iraq War, Russia-Ukraine war and the IranUS conflict can help. We can see if the change from trending to behavior is common, in conflicts that affect oil prices or just happened in this one case.

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